

ASSIGNMENT-10.2

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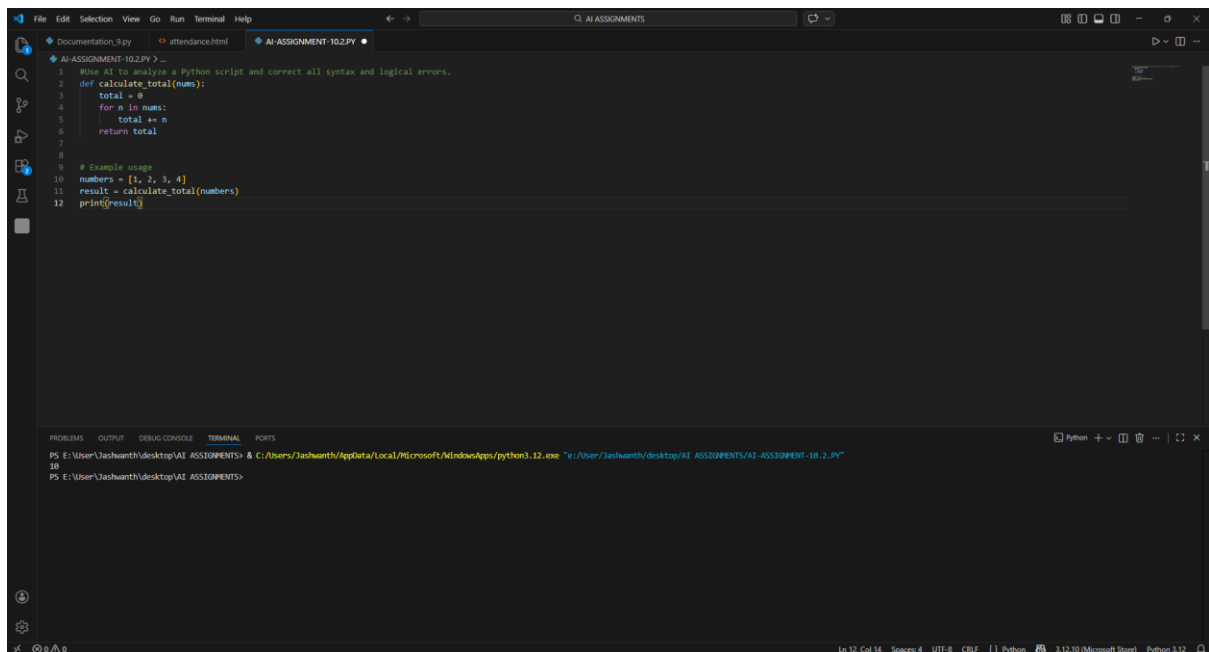
BATCH: 30

TASK-1

PROMPT:

Use AI to analyze a Python script and correct all syntax and logical errors.

CODE:



The screenshot shows a Visual Studio Code editor window with a file named 'AI-ASSIGNMENT-10.2.PY'. The code in the editor is as follows:

```
1 #Use AI to analyze a Python script and correct all syntax and logical errors.
2 def calculate_total(nums):
3     total = 0
4     for n in nums:
5         total += n
6     return total
7
8
9 # Sample usage
10 numbers = [1, 2, 3, 4]
11 result = calculate_total(numbers)
12 print(result)
```

Below the editor, the terminal window shows the command to run the script and its output:

```
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:/User/jashwanth/desktop/AI ASSIGNMENTS/AI-ASSIGNMENT-10.2.PY"
10
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS>
```

OBSERVATON:

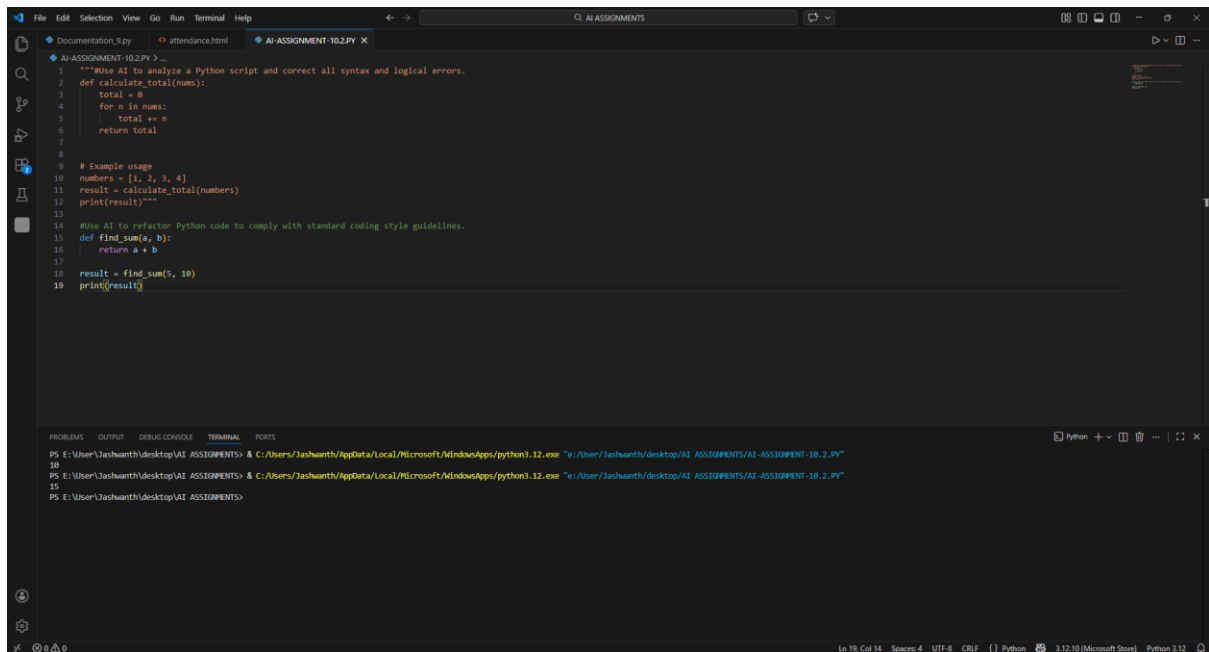
In this task, AI helped me find mistakes in the program. The code had missing colons and wrong indentation, which caused syntax errors. There was also a logical mistake where one variable name (sum) was used inside the function but a different variable (total) was returned. AI corrected these mistakes and made the program run successfully.

TASK-2

PROMPT:

Use AI to refactor Python code to comply with standard coding style guidelines.

CODE:



```
1 """Use AI to analyze a Python script and correct all syntax and logical errors.
2 def calculate_total(nums):
3     total = 0
4     for n in nums:
5         total += n
6     return total
7
8
9 # Example usage
10 numbers = [1, 2, 3, 4]
11 result = calculate_total(numbers)
12 print(result)"""
13
14 # Use AI to refactor Python code to comply with standard coding style guidelines.
15 def find_sum(a, b):
16     return a + b
17
18 result = find_sum(5, 10)
19 print(result)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:/User/jashwanth/desktop/AI ASSIGNMENTS/AI-ASSIGNMENT-10.2.PY"

10

PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:/User/jashwanth/desktop/AI ASSIGNMENTS/AI-ASSIGNMENT-10.2.PY"

15

PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS>

OBSERVATON:

In this task, AI improved the way the code was written. The function name was changed to follow proper naming rules (snake_case). Spacing between variables was corrected, and the code was formatted neatly. The program became more organized and easier to read.

TASK-3

PROMPT:

Use AI to improve code readability without changing its functionality.

CODE:

```
File Edit Selection View Go Run Terminal Help
AI-ASSIGNMENT-10.2.PY
11 def find_sum(a, b):
12     return a + b
13
14 result = find_sum(5, 10)
15 print(result)
16
17 #Use AI to improve code readability without changing its functionality.
18
19 def calculate_adjusted_difference(base_value, factor):
20     return base_value - (factor * 2)
21
22 result = calculate_adjusted_difference(10, 3)
23 print(result)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS E:\User\Jashwanth\Desktop\AI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:\User\Jashwanth\Desktop\AI ASSIGNMENTS\AI-ASSIGNMENT-10.2.PY"

10

PS E:\User\Jashwanth\Desktop\AI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:\User\Jashwanth\Desktop\AI ASSIGNMENTS\AI-ASSIGNMENT-10.2.PY"

15

PS E:\User\Jashwanth\Desktop\AI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:\User\Jashwanth\Desktop\AI ASSIGNMENTS\AI-ASSIGNMENT-10.2.PY"

4

PS E:\User\Jashwanth\Desktop\AI ASSIGNMENTS>

Ln 22, Col 14, Spaces: 4 UTF-8 CRLF Python 3.12.10 (Microsoft Store) Python 3.12

OBSERVATON:

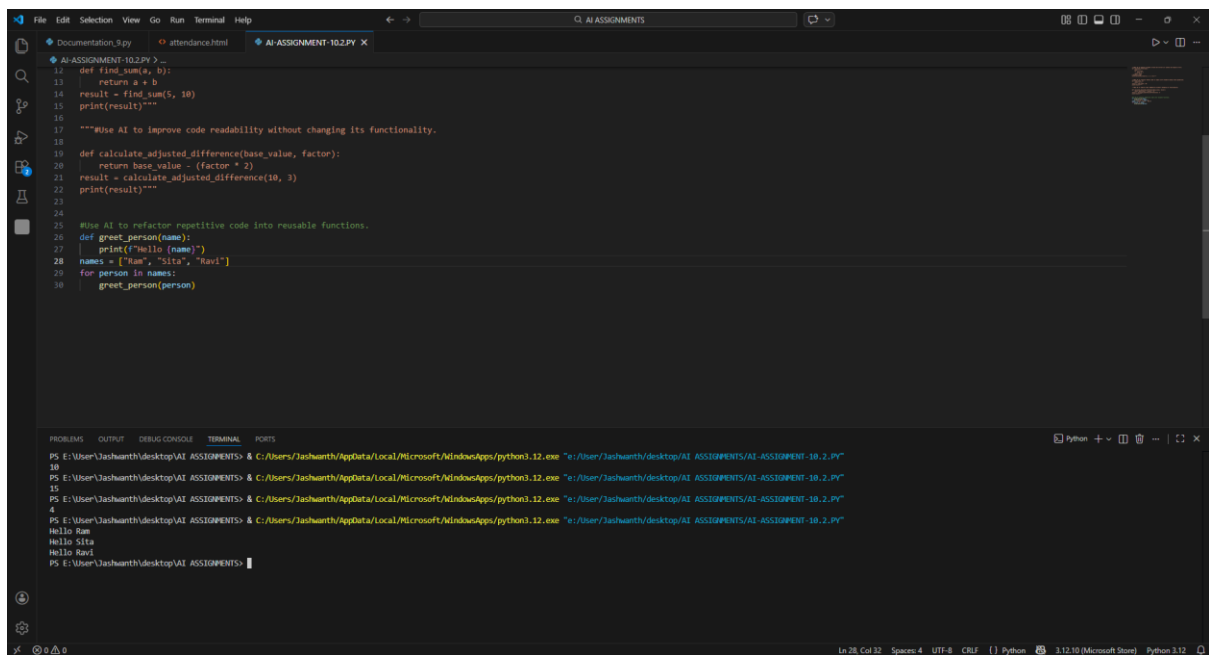
In this task, AI replaced unclear function and variable names with meaningful ones. It also corrected indentation and slightly modified the formatting of the mathematical expression to make it easier to understand. The functionality of the program remained the same.

TASK-4

PROMPT:

Use AI to refactor repetitive code into reusable functions.

CODE:



The screenshot shows a Visual Studio Code editor window with a Python file named 'AI-ASSIGNMENT-10.2.PY'. The code includes a function 'find_sum(a, b)' that returns the sum of two numbers, a docstring explaining its use, a function 'calculate_adjusted_difference(base_value, factor)' that returns the base value minus the product of the factor and base value, and a function 'greet_person(name)' that prints a greeting. A list of names is defined, and a loop calls 'greet_person' for each name. The terminal at the bottom shows the execution of the script, which outputs 'Hello Ram', 'Hello Sita', and 'Hello Ravi'.

```
11 def find_sum(a, b):
12     return a + b
13
14 result = find_sum(5, 10)
15 print(result)
16
17 """Use AI to improve code readability without changing its functionality.
18
19 def calculate_adjusted_difference(base_value, factor):
20     return base_value - (factor * 2)
21
22 result = calculate_adjusted_difference(10, 3)
23 print(result)
24
25 """Use AI to refactor repetitive code into reusable functions.
26
27 def greet_person(name):
28     print(f"Hello {name}")
29
30 names = ["Ram", "Sita", "Ravi"]
31 for person in names:
32     greet_person(person)
```

```
PS E:\User\Jashwanth\Desktop\AI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:/User/Jashwanth/Desktop/AI ASSIGNMENTS/AI-ASSIGNMENT-10.2.PY"
10
PS E:\User\Jashwanth\Desktop\AI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:/User/Jashwanth/Desktop/AI ASSIGNMENTS/AI-ASSIGNMENT-10.2.PY"
15
PS E:\User\Jashwanth\Desktop\AI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:/User/Jashwanth/Desktop/AI ASSIGNMENTS/AI-ASSIGNMENT-10.2.PY"
4
PS E:\User\Jashwanth\Desktop\AI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:/User/Jashwanth/Desktop/AI ASSIGNMENTS/AI-ASSIGNMENT-10.2.PY"
Hello Ram
Hello Sita
Hello Ravi
PS E:\User\Jashwanth\Desktop\AI ASSIGNMENTS>
```

OBSERVATON:

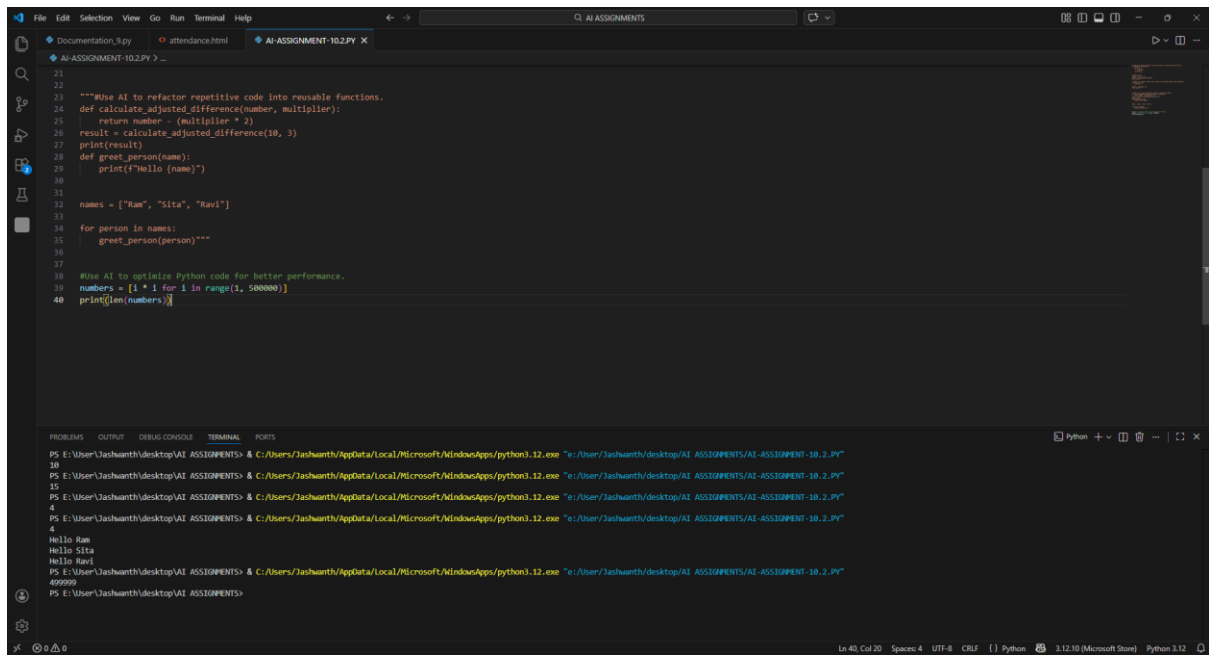
In this task, repetitive print statements were converted into a reusable function. Instead of writing similar lines multiple times, a function was created and called using a loop. This reduced repetition and made the program more structured.

TASK-5

PROMPT:

Use AI to optimize Python code for better performance.

CODE:



The screenshot shows a Visual Studio Code editor window with a Python file named 'AI-ASSIGNMENT-10.2.PY'. The code includes a function to calculate adjusted difference, a greet function, a list of names, and a list comprehension for generating numbers. The terminal at the bottom shows the execution of the script, which outputs 'Hello Ram', 'Hello Sita', 'Hello Ravi', and '400000'.

```
21
22
23 """Use AI to refactor repetitive code into reusable functions.
24 def calculate_adjusted_difference(number, multiplier):
25     return number - (multiplier * 2)
26 result = calculate_adjusted_difference(10, 3)
27 print(result)
28 def greet_person(name):
29     print(f"Hello {name}")
30
31 names = ["Ram", "Sita", "Ravi"]
32
33 for person in names:
34     greet_person(person)"""
35
36
37
38 #Use AI to optimize Python code for better performance.
39 numbers = [i * i for i in range(1, 500000)]
40 print(len(numbers))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS E:\User\Jashwanth\Desktop\VAI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:/user/jashwanth/desktop/AI ASSIGNMENTS/AI-ASSIGNMENT-10.2.PY"
10
PS E:\User\Jashwanth\Desktop\VAI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:/user/jashwanth/desktop/AI ASSIGNMENTS/AI-ASSIGNMENT-10.2.PY"
15
PS E:\User\Jashwanth\Desktop\VAI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:/user/jashwanth/desktop/AI ASSIGNMENTS/AI-ASSIGNMENT-10.2.PY"
4
PS E:\User\Jashwanth\Desktop\VAI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:/user/jashwanth/desktop/AI ASSIGNMENTS/AI-ASSIGNMENT-10.2.PY"
4
Hello Ram
Hello Sita
Hello Ravi
PS E:\User\Jashwanth\Desktop\VAI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:/user/jashwanth/desktop/AI ASSIGNMENTS/AI-ASSIGNMENT-10.2.PY"
400000
PS E:\User\Jashwanth\Desktop\VAI ASSIGNMENTS>
```

Ln 40, Col 20, Spaces: 4, UTF-8, CRLF, Python 3.12.10 (Microsoft Store), Python 3.12

OBSERVATON:

In this task, AI optimized the code by replacing the normal loop with a list comprehension. This made the code shorter and improved performance. The output remained the same, but the execution became faster and cleaner.